Project Report: Vehicle Parking App (NowPark)

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Description

This project aims to build a multi-user vehicle parking system where users can book or release parking spots, and admins can manage lots, view summaries, and control the system. It focuses on usability, dynamic spot management, and system transparency.

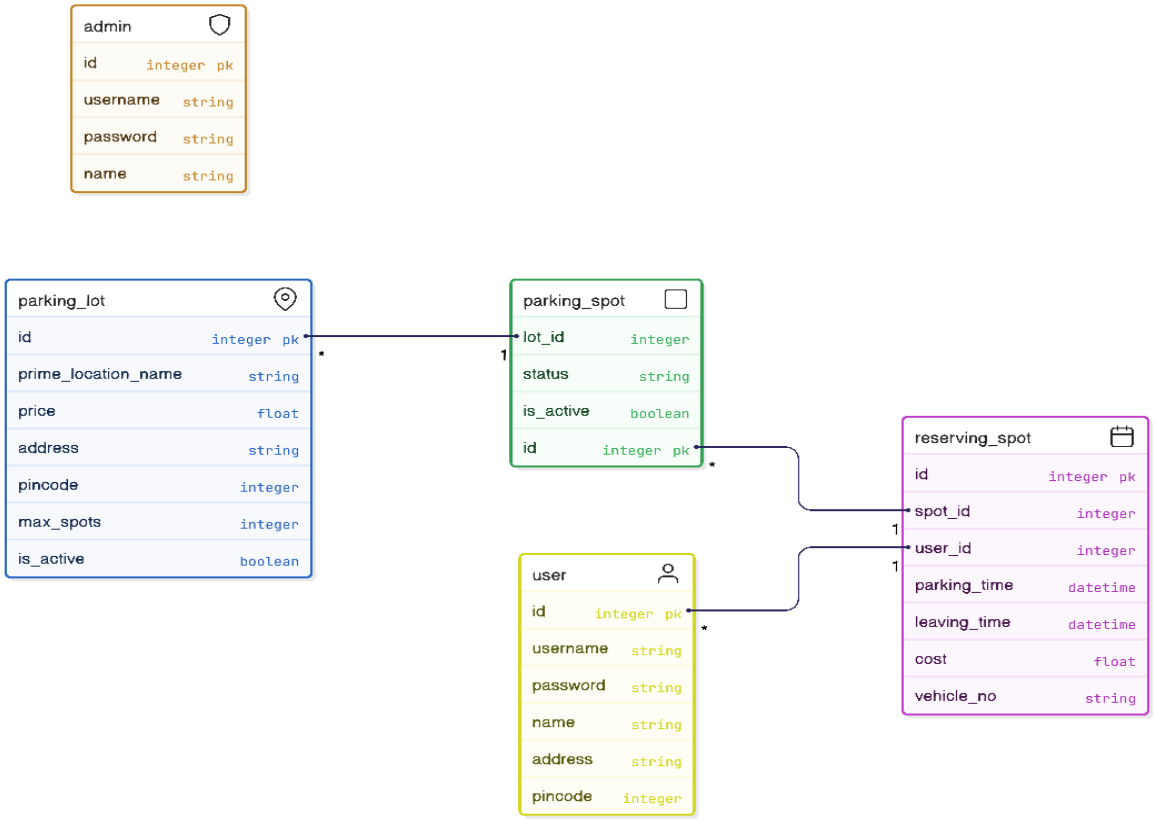
Technologies Used

- Flask
- Flask-SQLAlchemy
- Flask-Login
- Werkzeug
- SQLAlchemy
- SQLite
- Chart.js
- Bootstrap
- HTML, CSS, Jinja2

Flask provides a lightweight backend framework, SQLAlchemy handles ORM, Flask-Login manages user sessions, Chart.js is used for admin visualizations, and Bootstrap supports responsive UI design.

DB Schema Design

The database is organized by separating users, parking lots, parking spots, and reservations into distinct tables. This structure, supported by foreign keys, ensures data consistency, avoids duplication, and supports scalability by allowing efficient management of multiple users, lots, and bookings.



Architecture and Features

Implemented Features:

- User registration and login system with session management
- Auto-allocation of first available parking spot
- Booking history and real-time spot status for users
- Admin dashboard showing system usage and cost visualizations
- CRUD operations for lots and spots
- Secure password hashing
- Chart.js used to create dynamic bar charts for reservation summaries

Additional Features:

- Admin summary dashboard
- Cost calculation on completed reservations
- Responsive UI with Bootstrap
- Modular and clean project structure

```
VEHICLE-PARKING-APP/
│
├── App/
│   ├── controllers/
│   │   ├── admin.py
│   │   ├── main.py
│   │   └── user.py
│   │
│   ├── static/
│   │   ├── css/
│   │   └── images/
│   │
│   ├── templates/
│   │
│   ├── __init__.py
│   ├── decorators.py
│   └── models.py
│
├── instance/
│   └── site.db
│
├── run.py
├── requirements.txt
├── README.md
```

Video Link

https://drive.google.com/file/d/1rnylDZnfNDzgsxgW7-5nAD5OdsMP9fCs/view?usp=sharing